Roll No._____ Name_____

DEPARTMENT OF MATHEMATICS, T. I. E. T., PATIALA

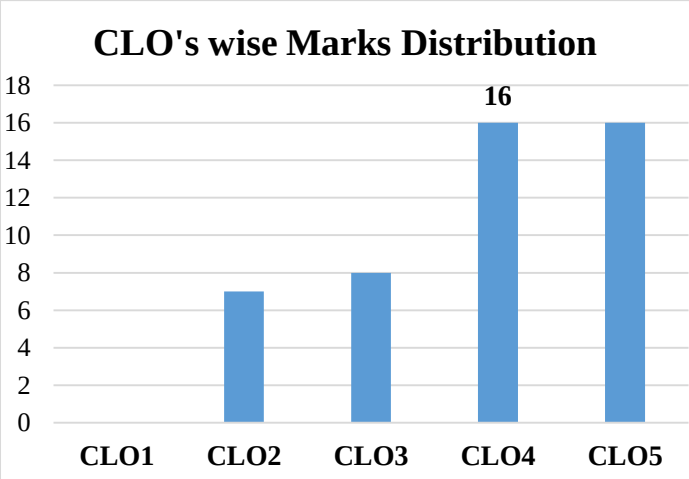
M.Sc. (Maths. and Computing) Session: 2023-24 (Even) End Sem. Examination, 2024

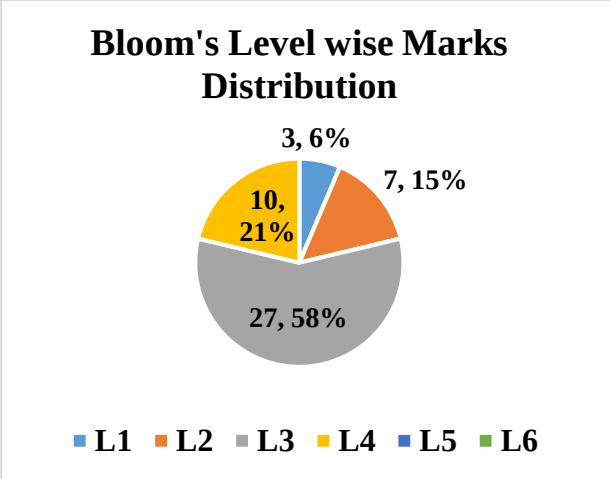
Time: 3 Hrs. Numerical Analysis (PMA-208) May 13, 2024 Max. Marks: 40

Note: Answer any five questions. Course Coordinator: Dr. Arvind Kumar Lal

1(a)	**Assume that $g \in C[a,b]$. If the range of the mapping $y = g(x)$ satisfies $a \le y \le b$ for all $a \le x \le b$. Furthermore, suppose that $g'(x)$ is defined over (a,b) and the positive constant $0 < \lambda < 1$ exists with $\lvert g'(x) \rvert \le \lambda < 1$ for all $x \in (a,b)$. Show that fixed point iterates $\mathbf{x_{n+1}} = \mathbf{g(x_n)}, \mathbf{n \ge 0}$ converges to a root α of $f(x) = 0$ in $[a,b]$ for any choice $\mathbf{x_0} \in [a,b]$. (4.0) [CO2, L2]**
(b)	**Working with four decimal digit rounding arithmetic, perform one iteration of the Newton's method to obtain the root of the following system of nonlinear equations: $x^2y + y^3 = 10$; $xy^2 - x^2 = 3$ using $\mathbf{x_0} = 0.8, \mathbf{y_0} = 2$.** **(3.0) [CO2, L3]**
2(a)	**Working with four decimal digit rounding arithmetic, perform one iteration of QR algorithm to find all the eigenvalue of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$. (5.0) [CO3 L3]**
(b)	**Working with four decimal digit rounding arithmetic, perform only four iterations of Power method to find the largest eigenvalue in magnitude and the corresponding eigenvector of the matrix $\mathbf{A} = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ with $\mathbf{X^{(0)}} = [\ 1,\ -1,\ 2\ ]^{\mathbf{T}}$.** **(3.0) [CO3, L3]**
3(a)	**Given that $f(1) = 1.5709, f(4) = 1.5727$ and $f(6) = 1.5751$. Working with four decimal digit rounding arithmetic, find Lagrange's interpolating polynomial $P_2(x)$ with these data. If we add one more point $f(7) = 1.5789$ in the above data, then find Newton's divided difference interpolating polynomial $P_3(x)$ in such a way that $P_3(x) = P_2(x) + r(x)$. Writing explicit form of $r(x)$, estimate $P_3(1.5)$.** **(5.0) [CO4, L4]**
(b)	**Find third order Hermite interpolation formula passing through points $(x_i, y_i, m_i), i = 0, 1$.** **(3.0) [CO4, L1]**
4(a)	**Suppose $x_0, x_1, \cdots, x_n$ are distinct numbers in the interval $[a,b]$ and $f \in C^{n+1}[a,b]$, Then prove that for each x in $[a,b]$, a number $\xi(x)$ (generally unknown) between $x_0, x_1, \cdots, x_n$, and hence in (a,b) exists with $f(x) = P(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x - x_0).(x - x_1) \cdots (x - x_n)$, where $P(x)$ is the Lagrange's interpolating polynomial.** **(3.0) [CO4, L2]**
(b)	**Construct natural cubic spline on the subinterval $[0, 0.25]$ that passes through $(0.25, 0.7071)$, $(0.5, 0.0)$, $(0.75, -0.7071)$ and $(1, -1)$ with the**

	boundary conditions $s_0 = 0$ and $s_4 = 0$. Hence, estimate $g(0.5)$. (5.0) [CO4, L4]
5 (a)	Find the quadrature formula $\int_0^1 \frac{f(x)}{\sqrt{x(1-x)}} dx = \alpha f(0) + \beta f(0.5)$ which is exact for polynomials of highest possible degree. (4.0) [CO5, L3]
(b)	Find the approximate value of the integral $\int_0^{\pi} sin\ x\ dx$ using composite Trapezoidal rule with two, three and five nodes and Romberg integration. Compare each approximate values with actual value. (4.0) [CO5, L3]
6(a)	Working with four decimal digit rounding arithmetic, find the approximate solution of the following boundary Value problem using finite difference method: $\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = x + 4,\ 0 \le x \le 0.6$ subject to boundary conditions: $y'(0) = 1,\ y(0.6) = 1.96$ with N = 3. (4.0) [CO5, L3]
(b)	Working with five decimal digit rounding arithmetic, solve the following problem: Consider an initial value problem $\frac{dy}{dx} = y + e^x, y(0) = 1$. For given value of $y(0.5) = 1.824$ and $y(1.0) = 3.718$, compute first $y(1.5)$ using Runge - Kutta fourth order method with step size $h = 0.5$ and then apply fourth order Adams - Bashforth - Moulton method to estimate $y(2.0)$. (4.0) [CO5, L3]





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